Online appendix

Benchmark corpora for detecting LLM-era bots

This appendix holds what the paper points to. Every number in it is reproduced by the code in this repository (`scripts/run_all.sh`) and checked against `results/expected/`.

1 How the ten corpora were found

The search ran on 20 August 2026 with web search and the OpenAlex API, after exploratory searches earlier that month. It followed three axes. First, keyword variants for how the field names these bots ("LLM-driven", "LLM-powered", "AI-powered" and "generative agent", each with "social bot", "detection", "benchmark" or "dataset"). Second, searches by dataset name, using each corpus already found as an entry point. Third, a forward citation chase through OpenAlex from two anchors, Fox8-23 (57 citing works at the time of the search) and BotSim-24 (12). The chase found the most; the keyword axis found the least, because the field's naming is unstable. Each axis stopped when a query returned nothing new. Which corpora a paper evaluates on was read from its full text, since abstract pages do not name them.

We kept every account-labelled corpus of social-media bots the search returned that either carries a published account-level detection result from a model that uses content or was released to study a campaign or information operation. Corpora that label texts rather than accounts, such as MultiSocial, are out of scope (Sect. 2.1 of the paper). We did not keep the exact wording of every keyword query or per-query screening counts, so this appendix records the procedure and its stopping rule, not a screening count.

2 The ten corpora

That search returns the ten corpora in Table 1 of the paper. Six have published account-level detection results from models that use content. The other four were built mainly to study campaigns and information operations rather than detection.

Most of the ten lack one of the three. *Three hold no LLM-written posts.* TwiBot-20 [3] contains accounts crawled outwards from seed users in politics, business, entertainment and sports. TwiBot-22 [4] uses a larger crawl of the same kind; both release the posts. Twitter IO [10] contains accounts Twitter removed as state-linked influence operations. All three were collected before ChatGPT's release in November 2022.

Three have no human class. CMV [5] releases 33 undisclosed LLM accounts that argued in a Reddit debate forum, all of them the experiment's bots [9]. Yard Signs [7] contains only accounts from a 2024 political campaign on X that replied to ordinary users, with no human comparison class. Masquerade-23 [6] is text from Chirper.ai, a platform whose users are all bots, so on its own it has no human class; we use it below in rebuilt form.

Two release no posts. MGTAB [11] contains expert-annotated accounts from a Twitter debate and releases extracted features rather than posts. Bluesky [8] contains 10,610 accounts from the platform Bluesky, with a label and source for each account but no posts.

For Table 1 of the paper, MGTAB, TwiBot-20 and TwiBot-22 each used one crawl and labelled the accounts afterwards. Bluesky is partly LLM-written: 460 of its 4,215 labelled bots are marked as generative. MGTAB releases embeddings rather than posts; Bluesky releases account labels and their sources.

3 Baselines

On Fox8-23 and BotSim-24 our runs use the accounts with at least 50 usable posts (20 characters or more), and score every account out of fold under five-fold cross-validation. We use cross-validation because neither corpus ships a split: the published results use a random held-out split that each paper draws in its own code. On Masquerade-rebuilt the published scores are for a three-way task that adds TwiBot-20's pre-LLM bots, so they are not directly comparable with our two-class result.

The simple baseline. Where the tables say "bag of words" the model is TF-IDF over the account's own posts with a minimum document frequency of two, followed by logistic regression, scored out of fold under five-fold cross-validation. The whole pipeline is fitted inside each training fold, and no two accounts share a document. The Fox8-23 and BotSim-24 bot-versus-human runs join each account's 50 messages; Sect. efapp:io gives the separate Twitter IO settings. The human-source comparisons on Fox8-23 join 20 messages, so these two analyses do not use the same amount of text.

Masquerade-rebuilt is scored on the text contrast it creates, Chirper.ai posts against TwiBot-20's human accounts, with 5,158 accounts per class.

We report the bag-of-words result as AUC everywhere except Sect. 2.2 of the paper, where we also compute F1 to compare with published F1 scores. The papers do not state their F1 thresholds; the released TRACE-Bot and BotDMM code takes the more probable class, a cut at 0.5, which we also use. In the 50-message condition, any threshold between the two classes gives the same result: the out-of-fold scores do not overlap, the lowest bot scoring 0.737 against the highest human 0.471 on Fox8-23 and 0.704 against 0.240 on BotSim-24, so the AUC is exactly 1 and every threshold between them gives the same F1. The bag of words also needs little text: 20 randomly selected posts concatenated and then truncated to 25 words in total still give F1 0.866 on Fox8-23 and 0.897 on BotSim-24. This truncated condition does overlap, and its F1 does depend on the cut. TwiBot-22's bag of words is scored on the 6,920 labelled accounts with at least one English reply to an English post that is not in a thread of their own; each account's replies are joined and cut to 20,000 characters.

The attribute baseline. A single account attribute uses the same five-fold procedure on one feature, so its direction is learned on the training folds rather than chosen with the labels. The attribute itself is picked by whichever scores best, and the shuffled-label null repeats that choice, which is why those nulls sit slightly above 0.5. For each pair of Fox8-23's human sources, we choose the best of seven attributes, such as the follower count or the length of the bio. On BotSim-24 the best attribute is whether the account has a bio, a non-empty profile description: all 1,000 generated accounts have one, against 278 of the 1,907 human accounts (14.6%).

Twitter IO is scored one campaign at a time, each with whichever attribute separates its own accounts best. Of its 27 campaigns, 25 have at least 20 accounts on each side and can be scored. Creation date gives the highest AUC in 21 of those 25; Table 1 of the paper gives the range of the best attribute's AUC across the scored campaigns. Above 10,000 accounts (TwiBot-20, MGTAB, TwiBot-22) the quantile transform behind the attribute baseline is fitted on a random subsample of 10,000 accounts; we fix its seed.

The Fox8-23 bag of words without the self-revealing phrases. The phrases are "as an AI language model" and seven variants, such as "I'm an AI", "OpenAI" and "ChatGPT"; no human account in the cohort uses any of them. On the same 500 bots and 500 humans, deleting the phrases from every message leaves AUC and F1 at 1.0000. Dropping the accounts whose 50 scored messages contain one keeps 222 bots, and the scores stay at 1.0000. Dropping those whose full released history contains one keeps 31 bots; AUC stays at 1.0000, and against 31 humans drawn at random, 100 times, F1 averages 0.9989. Shuffled labels give 0.47 to 0.51 in the first three runs.

Masquerade-rebuilt built as the detector papers describe it. Table 1 of the paper scores Masquerade-rebuilt as the text contrast it creates. Its published scores are for a different task: BotDMM, LLM-BotGuard and BotEvo classify humans, TwiBot-20's traditional bots and LLM-driven bots. To compare on that task, we built the corpus as BotDMM and LLM-BotGuard describe it, with our own draws where they are silent, and scored the same bag of words (Table 1). Against humans, the LLM-driven bots separate at F1 0.9900, as on Fox8-23 and BotSim-24. On the three-way task the same model reaches macro F1 0.8194, below the published 0.89 to 0.94. What it separates less well is TwiBot-20's traditional bots

construction	task and split	accuracy	macro F1	weighted F1	MCC	AUC	F1
BotDMM recipe	three-way, 70/20/10 test	0.8115	0.8194	0.8089	0.7084		
	three-way, five-fold CV	0.8150	0.8227	0.8117	0.7141		
	LLM-driven bots vs humans, 70/20/10 test	0.9806				0.9991	0.9730
	LLM-driven bots vs humans, five-fold CV	0.9927				0.9995	0.9900
	all bots vs humans, 70/20/10 test	0.8014				0.8939	0.8151
	all bots vs humans, five-fold CV	0.8082				0.8976	0.8219
LLM-BotGuard recipe	three-way, 70/20/10 test	0.8191	0.8024	0.8108	0.7224		
	three-way, five-fold CV	0.8316	0.8179	0.8251	0.7410		
	LLM-driven bots vs humans, 70/20/10 test	0.9886				0.9994	0.9857
	LLM-driven bots vs humans, five-fold CV	0.9926				0.9996	0.9907
	all bots vs humans, 70/20/10 test	0.8132				0.9068	0.8228
	all bots vs humans, five-fold CV	0.8238				0.9103	0.8342
published, BotDMM	three-way	0.9542	0.9362		0.9025		
published, LLM-BotGuard	three-way	0.8850	0.8851		0.8212		
published, LLM-BotGuard	bots vs humans	0.8817					
published, BotEvo	three-way		0.8930				

Table 1: Built by the detector papers' recipe, Masquerade-rebuilt still lets a bag of words separate LLM-driven bots from humans, but the bag of words falls below the published three-way scores. BotDMM replaces the tweets and profile description of 3,000 TwiBot-20 bot accounts with those of a Masquerade-23 (Chirper.ai) account and relabels them LLM-driven; LLM-BotGuard replaces the tweets of 3,500; BotEvo follows BotDMM. The papers do not state which accounts were drawn, how much of a timeline is replaced, or which F1 average is printed, so our rows are our own draws under their stated recipe, not their files. Our rows use TF-IDF over an account's words and logistic regression, fitted on the 70% split and scored on the 10% test split, or scored out of fold under five-fold cross-validation; the last column is the bot class's F1 at a 0.5 cut. The published rows are the papers' full models on their own draws.

from its humans (all bots against humans, AUC 0.8976), the same difficulty as Table 1's TwiBot-20 row. Table 1's attribute cell reads "accounts are TwiBot-20's" for the same reason: the LLM-driven accounts are TwiBot-20 bot accounts with replaced text. Under the BotDMM recipe and on Table 1's fields, the followers/friends ratio separates them from humans at 0.7495, against 0.7485 for all of TwiBot-20's bots, and no account count separates them from the traditional bots (0.48 to 0.51).

A bag of words fitted on one corpus and scored on the other. The transfer results use the text cohorts of Sect. 3, 500 bots and 500 humans for Fox8-23 and 329 and 408 for BotSim-24, 50 messages each joined into one document. The bag of words is fitted on every account of one corpus and scored on every account of the other, so no test account is seen in training and no cross-validation is needed. Fitted on Fox8-23 it scores 0.998 on BotSim-24, and fitted on BotSim-24 it scores 0.890 on Fox8-23; resampling the scored accounts 1,000 times puts these between 0.996 and 1.000 and between 0.869 and 0.909. Deleting "as an AI language model" from both corpora changes neither. Document length alone gives 0.781 and 0.744, and training on shuffled labels 0.51 and 0.49. Both sets of bots were written by OpenAI models, which may explain part of this transfer. In both corpora the humans were also gathered apart from the bots, so these transfer scores do not separate shared writing features from collection differences.

4 Collection routes

In Table 2, BotSim-24, Masquerade-rebuilt and Twitter IO each use a separate route for each class by construction. Bluesky records what brought each account in, such as a purchase or a community report; eight of its nine sources contain only one class. Meng et al. [8] report excluding features used in constructing their Bluesky dataset from their classifiers. **We measure how well the recorded collection route alone predicts the label**, Bluesky included (Table 2), and we require one route for both classes (Sect. 2.3 of the paper).

5 Every Twitter IO campaign, scored against its own controls

Twitter IO is the one family that releases its own controls, so each campaign is scored without us choosing the human side. Simple baselines still separate the classes: the median campaign has bag-of-words AUC

corpus, and recorded route	accounts	routes	single-class routes	route-only accuracy	majority rate	gain	V
Fox8-23, a source dataset	2,280	5	5 of 5	1.0000	0.5000	+0.5000	1.0000
BotSim-24, generated or scraped	2,907	2	2 of 2	1.0000	0.6560	+0.3440	1.0000
Masquerade-rebuilt, Chirper.ai or TwiBot-20	10,316	2	2 of 2	1.0000	0.5000	+0.5000	1.0000
Twitter IO, the archive or the control search	319,686	2	2 of 2	1.0000	0.9509	+0.0491	1.0000
Bluesky, a recorded source	10,610	9	8 of 9	0.9955	0.6027	+0.3927	0.9907
TwiBot-20, a seed domain	11,826	15	0 of 15	0.6553	0.5572	+0.0982	0.3712

Table 2: The collection route alone perfectly predicts the label in all three LLM-era candidates. A route is where an account comes from and what brings it into the corpus, such as a search, an earlier study or a community report. The table covers every corpus whose account routes are recorded or fixed by construction. Route-only accuracy is the accuracy of predicting the most common class within each route; majority rate is that of always predicting the largest class; gain is their difference. Cramér's V measures route–label association (0: independent; 1: route determines label). Only in TwiBot-20 does every route contain both classes. In BotSim-24, Masquerade-rebuilt and Twitter IO the route is the class by construction, so their accuracy of 1 is true by definition, not measured.

0.9665 (Table 3). Creation date is the best single attribute in 21 of the 25 scored campaigns, with median AUC 0.91. The creation-date result has two possible explanations, and the release does not separate them. Campaign accounts may really be registered in bursts, which is behaviour a detector should use, or the difference may come from how the controls were gathered, since they were chosen by hashtag and day and not by creation date.

How the two classes were gathered. The operation's accounts come from Twitter's archive with their whole timelines. The controls are accounts that used one of the operation's hashtags on the same day, with at most 100 of their posts from those days, and the authors note that a control's posts cover a short period while an operation account's can span years [10]. Across all 27 campaigns, the operation's accounts have more posts than their controls in 26, at a median ratio of 7.0. Their posts span a median of 137 days, against 0 for controls, and the most common language differs between the classes in 12 campaigns.

Scoring. Table 3 scores the 25 campaigns with enough accounts in both classes. The account-level baseline uses up to 200 posts per account, drops reposts, keeps messages of at least 20 characters without a language filter, caps each class at 4,000 accounts, and uses five-fold cross-validation. Bangladesh (11 accounts) and Russia_3 (5) fall below the floor of 20 accounts per class and are not scored.

Matching the two classes exactly by hashtag and day. The rule both classes share is a post under a given hashtag on a given day, so the match is made per message (Table 4). All four rows use posts in the operation's most common language, at most five posts per account, and the same number of posts per class. The first row draws posts at random. The second keeps only posts under a hashtag and day that both classes used. The third gives both classes exactly the same number of posts under each hashtag and day, and the fourth also deletes hashtags, mentions and URLs from the text and drops any text that occurs more than once. Folds are grouped by account, the AUC is computed inside each fold and averaged, and each cell is the mean of ten draws. Four of the 27 campaigns keep at least 100 exactly matched posts from at least 20 accounts in each class in all ten draws, and only those are scored. One Ecuador account appears in both classes; all of its posts are excluded. Shuffling the labels by account gives 0.49 to 0.51 in every cell.

Matching frequencies matters. An account-level check using three posts per account matched which hashtags and days appeared, but not how often. AUC fell in 2 of 10 campaigns and rose at the median, from 0.873 to 0.911. The exact message-level match controls those frequencies in both classes.

In the last two rows of Table 4 every hashtag and day holds the same number of bot and control posts in every draw.

6 Collecting a corpus, in full

The paper's Sect. 5 gives the short form of this advice; this is the long form it is shortened from.

campaign	bag of words	length	null	accounts, campaign / control
Qatar	0.8087	0.7127	0.5065	28 / 4,000
China_2	0.8698	0.8273	0.5214	179 / 4,000
Russia_4	0.9033	0.7987	0.5041	24 / 4,000
Venezuela_2	0.9058	0.9737	0.5187	33 / 3,002
Russia_2	0.9143	0.6644	0.4814	351 / 1,892
Thailand	0.9177	0.5363	0.5285	301 / 1,386
Iran_2	0.9491	0.6437	0.4915	289 / 4,000
Iran_3	0.9535	0.7137	0.5224	135 / 1,659
China_1	0.9548	0.7130	0.4948	683 / 4,000
Russia_5	0.9556	0.7578	0.4683	48 / 4,000
Cuba_2	0.9605	0.8255	0.5144	422 / 528
Ghana/Nigeria	0.9607	0.8685	0.4352	57 / 885
Ecuador	0.9665	0.6862	0.4965	768 / 4,000
Iran_1	0.9666	0.6985	0.4951	644 / 4,000
Spain	0.9672	0.8250	0.4761	210 / 1,067
Iran_6	0.9673	0.8774	0.4849	206 / 4,000
Catalonia	0.9686	0.6565	0.5458	47 / 1,723
UAE	0.9758	0.8196	0.4961	3,432 / 4,000
Egypt/UAE	0.9759	0.8461	0.4709	236 / 285
Russia_1	0.9763	0.7297	0.5088	3,123 / 4,000
Cuba_1	0.9800	0.8852	0.4871	377 / 4,000
Iran_4	0.9821	0.6075	0.4899	2,468 / 4,000
Armenia	0.9837	0.7253	0.5121	31 / 1,384
Venezuela_1	0.9928	0.9136	0.5005	577 / 4,000
Iran_5	0.9938	0.3054	0.5505	104 / 889

Table 3: A bag of words exceeds AUC 0.90 in 23 of 25 campaigns with no LLM-written posts. Each Twitter IO campaign is compared with its released controls at account level, using five-fold cross-validation (median AUC 0.9665; range 0.8087–0.9938). "Length" uses only the character count of joined posts; "null" uses shuffled labels. Below-chance AUCs retain their direction.

Where to get it. Most corpora come from a platform's API or from scraping. Both see only what is public and still online: the platform's own records stay out of reach, and removed accounts are gone. The EU's Digital Services Act gives researchers another way to obtain data [2]. Under its Article 40, a researcher can ask a very large online platform for data to study systemic risks. Bot activity is one of them: Article 34(2) names "inauthentic use or automated exploitation of the service". The request can cover public data (Article 40(12)) and, for vetted researchers, non-public data (Article 40(4), with access procedures in force since October 2025 [1]). In either case a researcher can ask for bots and humans collected by one route, and for the records needed to check it. What a platform hands over is settled case by case.

What to keep. We recommend keeping everything the platform's terms and the study's ethics approval allow. A post not kept now may be gone later, and collecting only what is still online introduces survivorship bias. Three things are easy to lose and hard to reconstruct:

- **Whole account histories.** As many posts per account as possible, for both classes, let an evaluation compare a few posts with a longer history. An analysis can subsample later, but it cannot recover posts that were never kept.
- **Each account's route.** A record of how every account entered the corpus, including any search terms used to find it, lets researchers check whether the route predicts the label and test a detector on accounts from a route held out of training (Sect. 2.3 of the paper).
- **The post each reply answers.** Its identifier and text let replies to the same posts be collected by one rule for both classes, which gives reply bots a shared route (Sect. 2.3 of the paper). It also lets future detectors use the conversation, not the reply alone.

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posts compared	Cuba_1	Ecuador	Russia_1	Spain
drawn at random	0.710	0.819	0.783	0.867
hashtag and day used by both	0.655	0.844	0.815	0.872
same number per hashtag and day	0.471	0.673	0.569	0.683
and no tags, no repeated text	0.495	0.632	0.581	0.612
posts per class	158	307	195	137

Table 4: Recovering a shared route lowers message-level AUC in all four Twitter IO campaigns. Median bag-of-words AUC falls from 0.801 to 0.621 when bots and controls contribute equal numbers of posts per shared hashtag and day. Rows progress from random posts to shared hashtag–day pairs, then equal class counts within each pair; the final row also removes hashtags, mentions, URLs and repeated text. AUCs average ten draws with cross-validation folds grouped by account.

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